

SAP CPI Project Documentation

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Scenario: Product Shipment Orders Processing

1.Scenario Overview

This integration handles **bulk shipment orders** by splitting them into individual orders, validating quantities, routing them based on region, and converting them into structured formats for warehouse systems. The scenario demonstrates the use of **Splitter**, **Filter**, **Router**, and **Converter** palette steps in SAP CPI and helps streamline logistics automation.

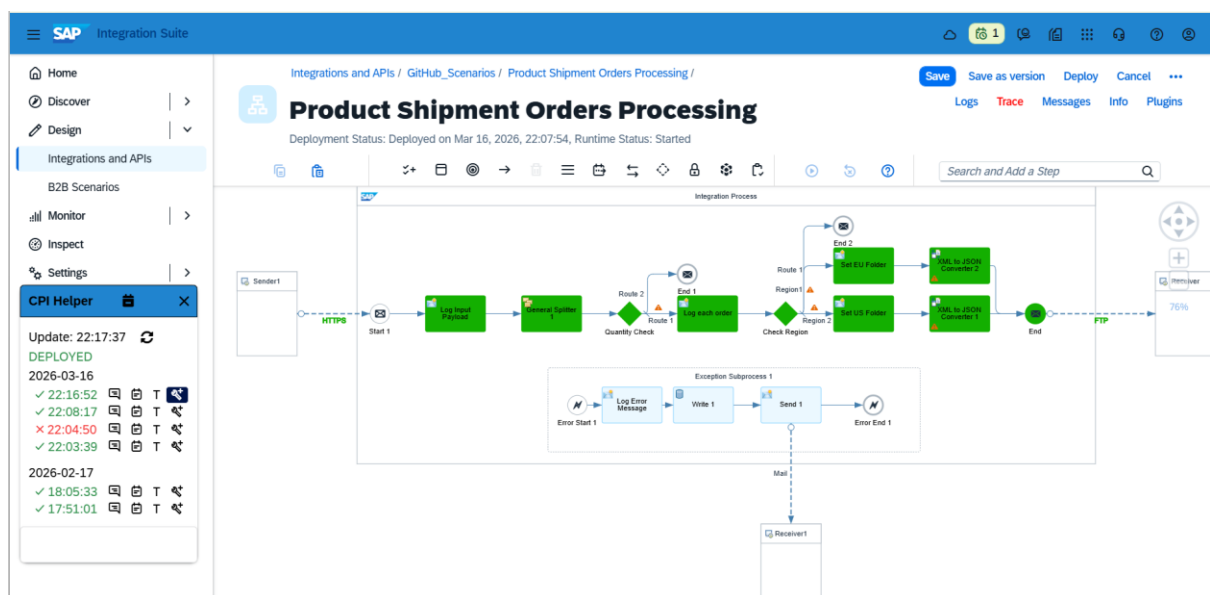
Business Use Case

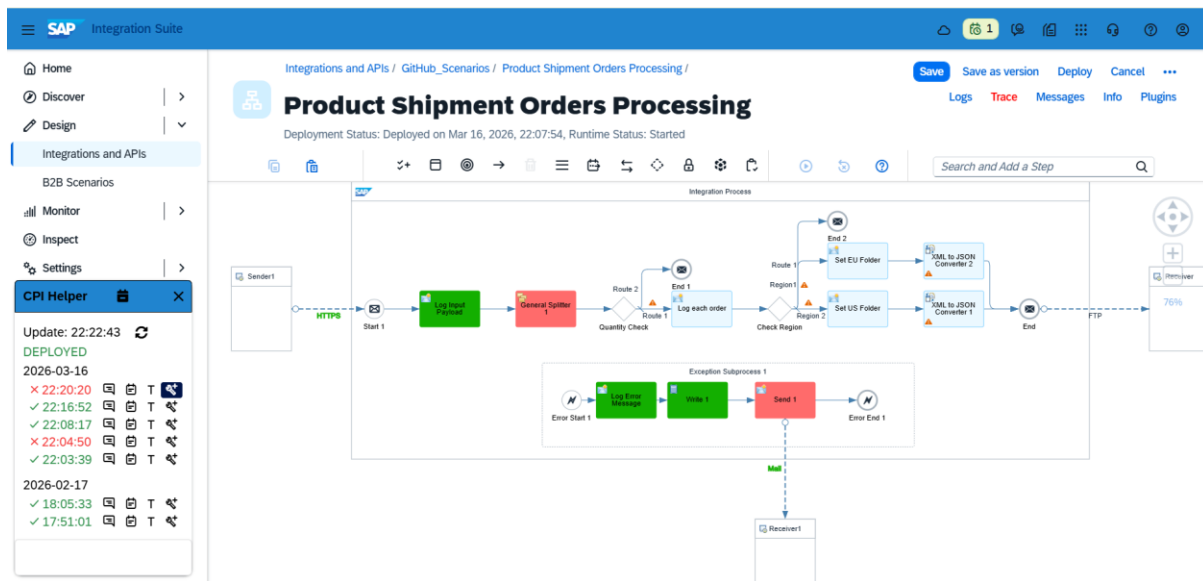
A logistics system sends a **daily shipment file** containing multiple product orders for different regional warehouses (EU, US). Each order includes the product ID, quantity, and region.

Processing needs to:

- Validate orders (quantity > 0)
- Route based on region
- Convert format from XML to JSON
- Store orders as separate files in **region-specific folders**

2.Architechture Diagram





3. Integration Flow

Steps

1. HTTP picks up batch XML file

HTTPS

General Connection Conditions

REQUEST PROCESSING

Address: /ProductShipment/Orders

Authorization: User Role

User Role: ESBMessaging.send

CSRF Protected: ☒

2. XML Splitter splits each <Order> from the batch
3. Router removes invalid orders (e.g., quantity = 0)
4. Content Modifier enriches each message with metadata (e.g., processed timestamp)

5. **Router** directs orders to regional branches (EU/US)

Router

General Processing

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	Route 2		☒
2	Route 1	//Quantity > '0'	☐

Router

General Processing

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	Region 2	//Region = 'US'	☐
2	Region1	//Region = 'EU'	☐
3	Route 1		☒

6. **XML to JSON Converter** transforms each valid order

7. **File Receiver Adapter** writes to /shipments/EU/ or /shipments/US/

Integration Process

FTP

General Target Processing

FILE ACCESS PARAMETERS

Directory: /shipments/\${property.Region}

File Name: \${property.ID}.json

Append Timestamp: ☐

CONNECTION PARAMETERS

Address: ftp.drivehq.com

Proxy Type: Internet

Encryption: Explicit FTPS

Credential Name: FTP_Connect

Timeout (in ms): 10000

Maximum Reconnect: 3

8. **Sample Input XML (Batch Shipment File)**

<Shipments>

<Order>

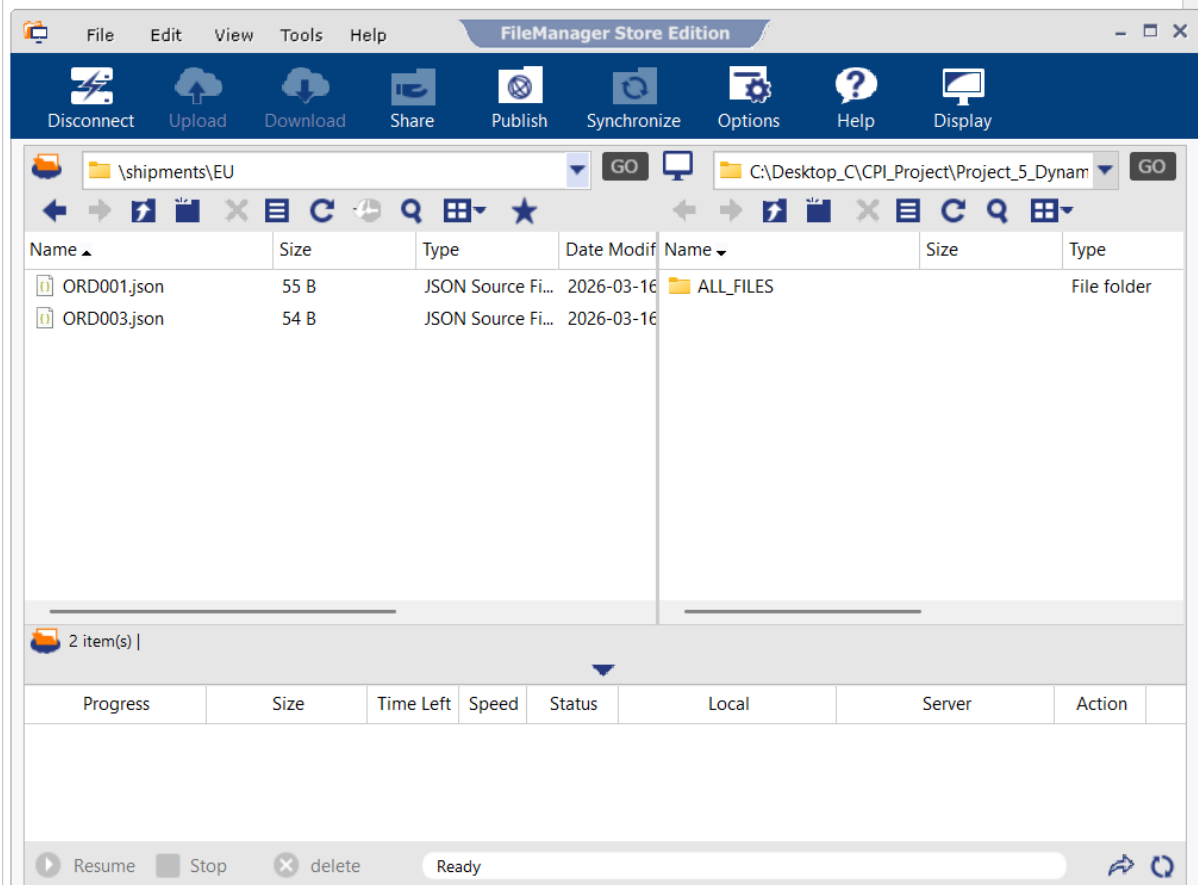
```
<ID>ORD001</ID>
  <Region>EU</Region>
  <Quantity>10</Quantity>
</Order>
<Order>
  <ID>ORD002</ID>
  <Region>US</Region>
  <Quantity>0</Quantity>
</Order>
<Order>
  <ID>ORD003</ID>
  <Region>EU</Region>
  <Quantity>5</Quantity>
</Order>
</Shipments>
```

Processing Example

- **Filtered out:** ORD002 (Quantity = 0)
- **Processed and routed:** ORD001 → EU folder, ORD003 → EU folder

9. JSON Output Example (ORD001)

```
{
  "Order": {
    "ID": "ORD001",
    "Region": "EU",
    "Quantity": 10
  }
}
```



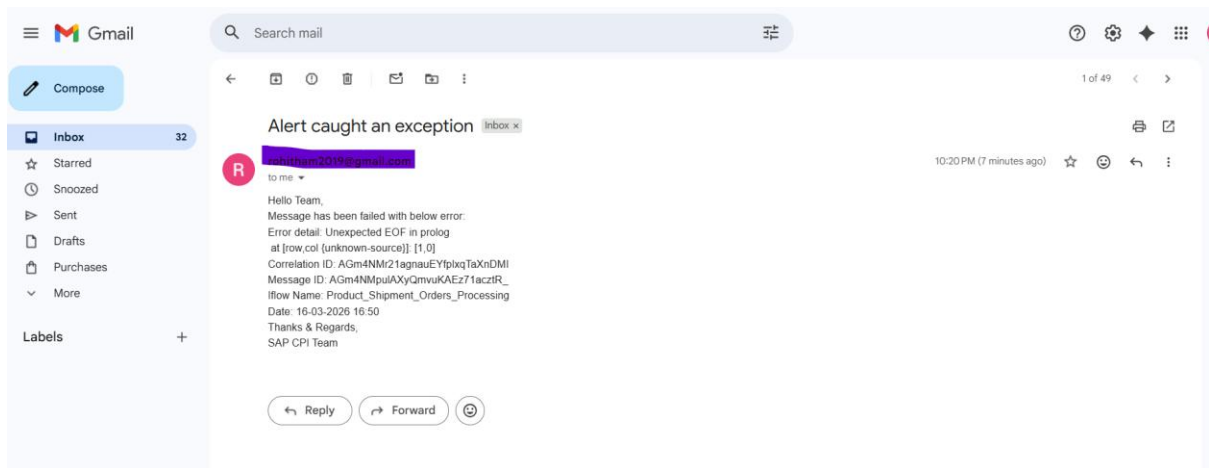
Files stored in HR system

File Output (File Receiver Adapter)

- Output path: `/shipments/EU/ORD001.json` `/shipments/EU/ORD003.json`
- Output format: JSON
- File naming: Based on order ID and region

10. Exception Handling

- **Catch step errors** in file write or transformation
- Logged to **Data Store**
- Notify logistics support team via **Gmail Adapter**



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Iflow Package



Product Shipment Orders Processing.zip